

RAG and multimodal systems

- Text RAG Pipeline
- Demo Session (1 Hour)

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Agenda

- • What is RAG?
- • Pipeline Overview
- • Chunking Strategies
- • Retriever
- • Router
- • Generator
- • Evaluation Metrics

What is RAG? (5 mins)

- Retrieval-Augmented Generation
- Combines retrieval + LLM generation
- Improves factual accuracy

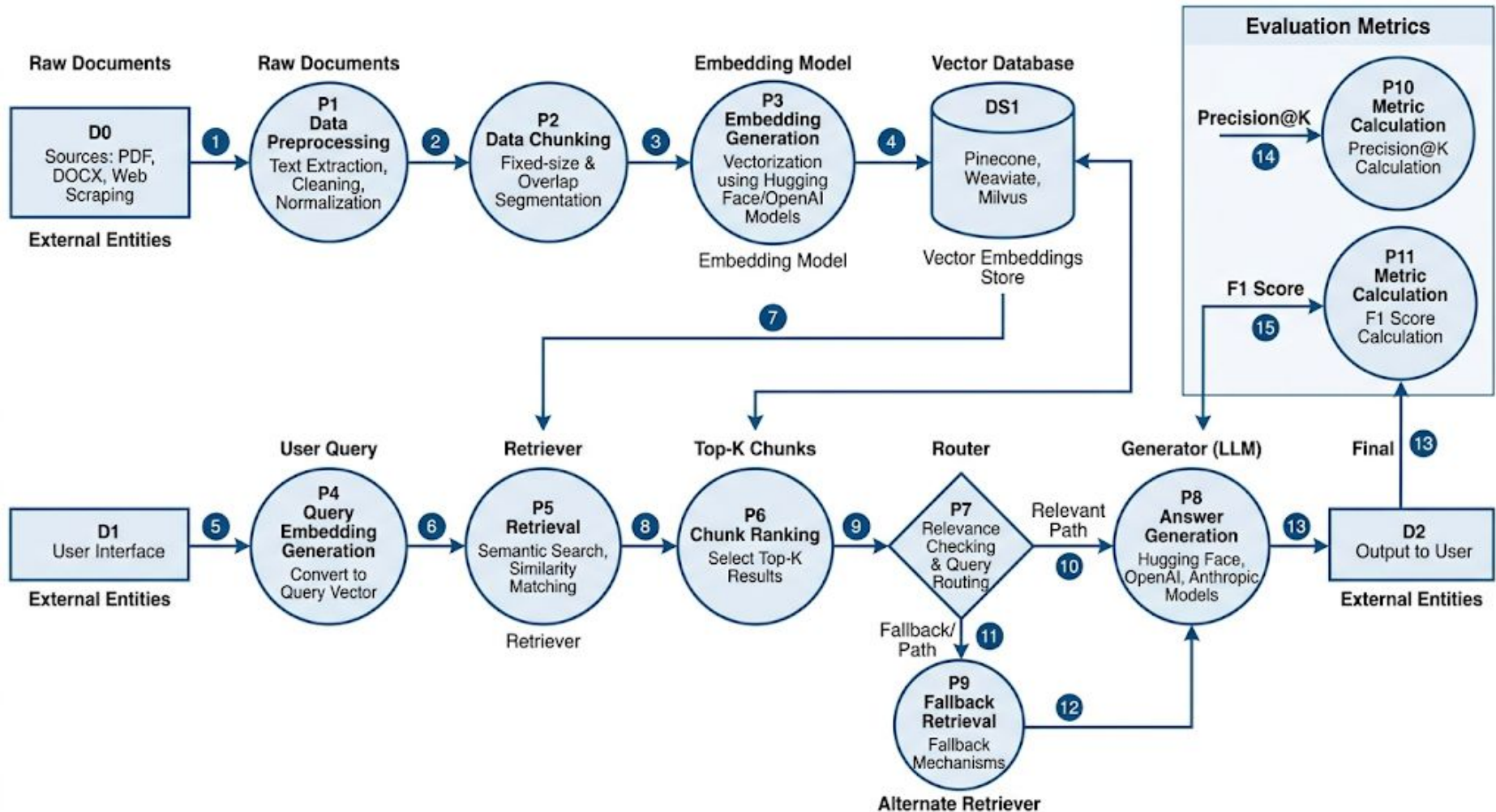
What is RAG?

- Retrieval-Augmented Generation (RAG) is a technique that enhances LLMs by adding an external knowledge retrieval step.
- It first retrieves relevant information from a data source (e.g., PDFs, databases, vector stores).
- The retrieved context is then passed to the LLM to generate more informed and grounded responses.
- This approach significantly improves factual accuracy and reduces hallucinations compared to standalone LLMs.

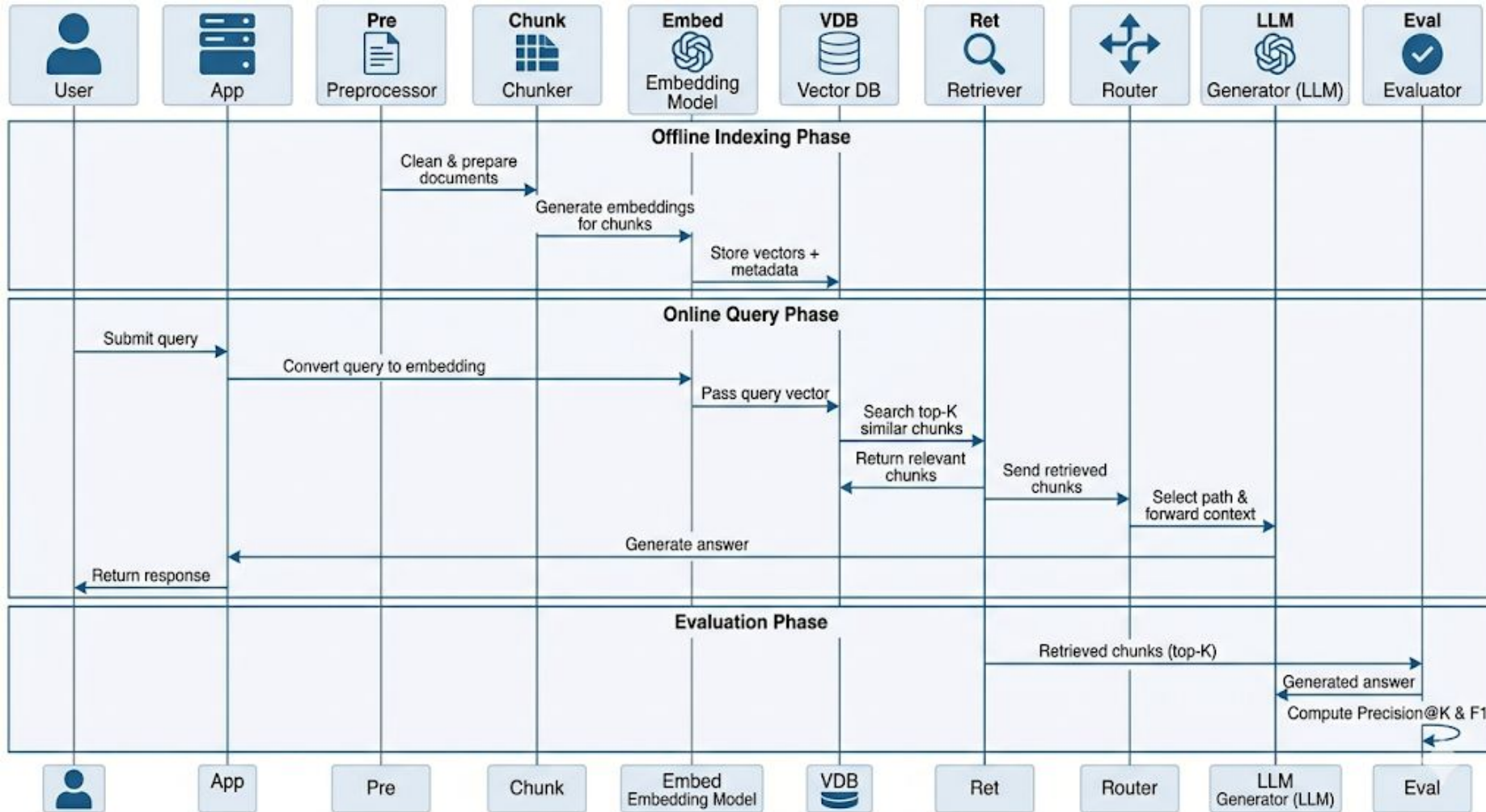
RAG Pipeline Overview (10 mins)

- 1. Document ingestion
- 2. Chunking
- 3. Embeddings
- 4. Retrieval
- 5. Generation

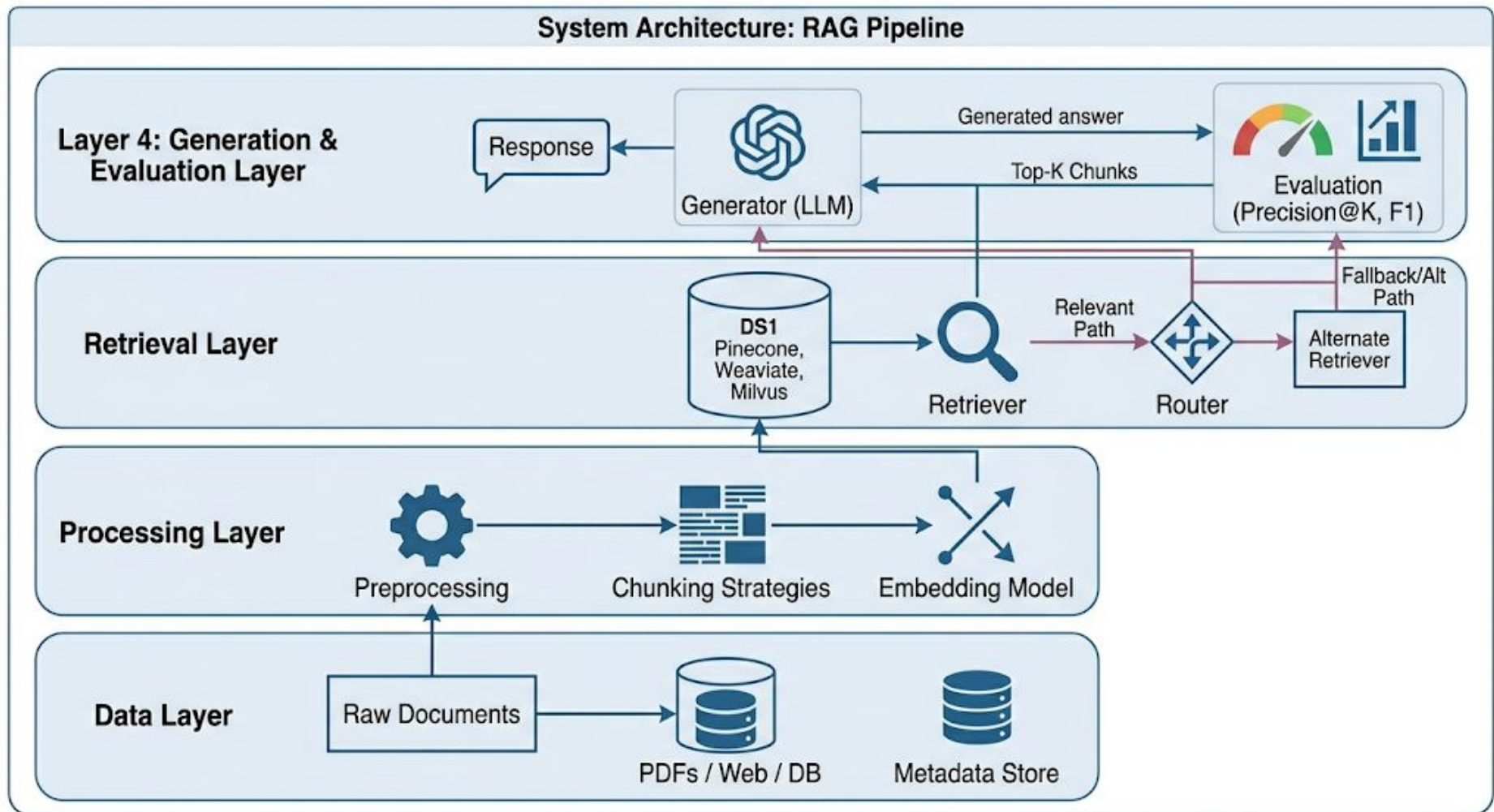
Data Flow Diagram



Sequence Diagram



4 Layer Architecture Diagram



Chunking Basics (10 mins)

- Splitting documents into smaller pieces
- Improves retrieval relevance
- Tradeoff: context vs precision

Chunking Basics

- Chunking is the process of splitting large documents into smaller, manageable pieces for processing.
- Smaller chunks help the retriever find more precise and relevant information for a query.
- However, very small chunks may lose context, while larger chunks may include irrelevant details.
- The key tradeoff is balancing context (completeness) vs precision (relevance) for optimal retrieval performance.

Chunking Strategies

- • Fixed-size chunking
- • Sliding window
- • Semantic chunking

Retriever (5 mins)

- Fetch relevant chunks
- Types:
 - • Dense (vector search)
 - • Sparse (BM25)
 - • Hybrid

Router

- Selects best retriever or path
- Used in multi-index systems
- Example: FAQ vs Docs routing

Router

- A router decides the best retrieval path or retriever based on the incoming query.
- It is commonly used in multi-index systems to direct queries to the most relevant data source.
- For example, it can route simple queries to an FAQ index and complex ones to detailed documents.
- This improves efficiency, relevance, and overall answer quality in RAG systems.

Generator

- LLM generates answer using retrieved context
- Prompt design is critical

Evaluation Overview (5 mins)

- Measure RAG performance
- Focus: retrieval + generation quality

Precision@K

- Fraction of relevant docs in top K
- Higher = better retrieval
- groundedness and relevance scores are useful!

F1 Score

- Harmonic mean of precision and recall
- Balances false positives & negatives

Hypothesis (10 mins)

1. **Smarter chunking improves retrieval relevance**

Using semantic or recursive chunking instead of fixed-size chunks will increase contextual coherence and retrieval quality.

2. **Hybrid retrieval outperforms single-method retrieval**

Combining dense (embeddings) + sparse (BM25) retrieval will capture both semantic meaning and keyword matches.

3. **Routing improves answer accuracy in multi-domain PDFs**

A router that directs queries to specialized indexes (e.g., FAQ vs technical sections) will reduce irrelevant context.

4. **Better prompt + context formatting improves generation quality**

Structuring retrieved chunks (ordering, deduplication, citation-style prompts) will improve LLM answer correctness.

Outcomes

1. **Higher Precision@K (retrieval quality)**
More relevant chunks appear in top-K results.
2. **Improved F1 score (end-to-end quality)**
Better balance of precision and recall in generated answers.
3. **Reduced hallucinations**
Answers are grounded in retrieved PDF content with fewer fabricated details.
4. **Better user satisfaction / answer usefulness**
Users find answers more accurate, complete, and context-aware.

How to test

1. Create a Gold Dataset

- Prepare ~50–100 question–answer pairs from the PDF
- Include:
 - Fact-based questions
 - Multi-hop questions
 - Edge cases

2. Retrieval Evaluation

- For each query:
 - Retrieve top-K chunks
 - Check if relevant chunk exists in top-K
- Metrics:
 - **Precision@K**
 - (Optional) Recall@K

How to test

3. Generation Evaluation

- Compare generated answer vs ground truth
- Metrics:
 - F1 score
 - Exact match (for factual Qs)

4. Ablation Testing

Test variations:

- Chunking:
 - Fixed vs Semantic vs Sliding window
- Retrieval:
 - Dense vs BM25 vs Hybrid

How to test

- With vs Without Router
- Prompt:
 - Basic vs structured prompt

👉 Measure impact of each on:

- Precision@K
- F1 score

5. Qualitative Evaluation

- Manually inspect:
 - Hallucinations
 - Missing context
 - Redundant chunks
 - Answer coherence

Hands-on RAG (10 mins)

- Google Colab!

Wrap-up + Q&A (5 mins)

- • RAG improves LLMs
- • Chunking is key
- • Retrieval quality matters
- • Evaluate properly